

Package Inserts		body weight corresponds to 0.1 ml of ENROLIN-LD 100MG/ML ORAL SOLUTION per kg body weight; the following calculation should be made to provide the required amount of ENROLIN-LD 100MG/ML ORAL SOLUTION per litre of drinking water:	
I.	Brand or Product Name: ENROLIN-LD 100MG/ML ORAL SOLUTION		
II.	Name and Strength of Active Substance: Enrofloxacin Base, 100mg in 1mL		
III.	Product Description: Clear, yellow solution.	ml ENROLIN-LD 100MG/ML ORAL SOLUTION per kg body weight daily	Average body weight (kg) of the animals to be treated
IV.	Target Species: Broilers, broiler breeders, replacement chickens, turkeys.	Total water consumption (l) of the flock at the previous day = ml ENROLIN-LD 100MG/ML ORAL SOLUTION per litre drinking water Care should be taken that the intended dose is completely ingested. The product may be put directly into the header tank or introduced via a water proportioner pump. For the preparation of the medicated water for small groups of animals the measuring cup included with the packaging should be used. During preparation of medicated water, the product should be admixed into water and not the other way round. Only sufficient medicated drinking water should be prepared to cover the daily requirements. Medicated water should be prepared every day, immediately prior to provision. Medication of the water supply should be continuous during the treatment period and no other source of water should be available. Medicated drinking water should be replaced every 24 hours.	
V.	Pharmacodynamics: Enrofloxacin is a synthetic, broad spectrum antimicrobial, bactericidal in action and effective against a wide range of gram positive and gram negative bacteria as well as mycoplasmas. It inhibits the enzyme DNA gyrase in the cell nuclei during replication of bacterial DNA. It also acts on bacterial cells during stationary phase by changing the permeability in the phospholipid cellular membranes. Antibacterial spectrum: Enrofloxacin is active against many Gram-negative bacteria, against Gram-positive bacteria and <i>Mycoplasma</i> spp. In vitro susceptibility has been shown in strains of (i) Gram-negative species such as <i>Pasteurella multocida</i> and <i>Avibacterium (Haemophilus) paragallinarum</i> and (ii) <i>Mycoplasma gallisepticum</i> and <i>Mycoplasma synoviae</i> . Types and mechanisms of resistance: Resistance to fluoroquinolones has been reported to arise from five sources, (i) point mutations in the genes encoding for DNA gyrase and/or topoisomerase IV leading to alterations of the respective enzyme, (ii) alterations of drug permeability in Gram-negative bacteria, (iii) efflux mechanisms, (iv) plasmid mediated resistance and (v) gyrase protecting proteins. All mechanisms lead to a reduced susceptibility of the bacteria to fluoroquinolones. Cross-resistance within the fluoroquinolone class of antimicrobials is common.	IX.	Contraindication: Do not use for prophylaxis. Do not use when resistance / cross resistance to (fluoro) quinolones is known to occur in the flock intended for treatment. Infections caused by <i>Streptococcus</i> spp., because of only marginal susceptibility to enrofloxacin. Do not use in other animals. Do not use in case of hypersensitivity to the active substances or any of the excipients.
VI.	Pharmacokinetics: The pharmacokinetics of enrofloxacin are such that both oral and parenteral administration lead to similar serum levels. Enrofloxacin possesses a high distribution volume. Tissue levels 2-3 times higher than that found in the serum have been demonstrated in laboratory animals and target species. Organs in which high levels can be expected are the lungs, liver, kidney, bone and lymphatic system. Enrofloxacin also distributes into the cerebrospinal fluid, the aqueous humour.	X.	Warning and Precautions: <u>Special warnings for each target species:</u> Treatment of <i>Mycoplasma</i> spp infections may not eradicate the organism. <u>Special precautions for use</u> <u>Special precautions for use in animals:</u> Official and local antimicrobial policies should be taken into account when the product is used. Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials. Whenever possible, fluoroquinolones should only be used based on susceptibility testing. Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance. Since enrofloxacin was first authorized for use in poultry, there has been widespread reduction in susceptibility of <i>E. coli</i> to fluoroquinolones and emergence of resistant organisms. Resistance has also been reported in <i>Mycoplasma synoviae</i> in the EU. As enrofloxacin will be partly excreted via kidneys, elimination will be delayed in cases with kidney disorders. <u>Special precautions to be taken by the person administering the veterinary medicinal product to animals:</u> Personal protective equipment consisting of impervious gloves should be worn when handling the veterinary medicinal product. Wash any splashes from skin or eyes immediately with water. Wash hands and exposed skin after use. Do not eat, drink or smoke while using the product. <u>Other precautions:</u> After the end of treatment, the watering system should be cleaned
VII.	Indication: Treatment of infections caused by the following bacteria susceptible to enrofloxacin: Chickens: <i>Mycoplasma gallisepticum</i> , <i>Mycoplasma synoviae</i> , <i>Avibacterium paragallinarum</i> , <i>Pasteurella multocida</i> . Turkey: <i>Mycoplasma gallisepticum</i> , <i>Mycoplasma synoviae</i> , <i>Pasteurella multocida</i> .		
VIII.	Recommended Dose and Administration: Chickens and turkeys: 10 mg enrofloxacin/kg bodyweight per day for 3–5 consecutive days. Treatment for 3–5 consecutive days; for 5 consecutive days in mixed infections and chronic progressive forms. If no clinical improvement is achieved within 2–3 days, alternative antimicrobial therapy should be considered based on susceptibility testing. The uptake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of ENROLIN-LD 100MG/ML ORAL SOLUTION should be adjusted accordingly. Taking into consideration that 10 mg enrofloxacin per kg		
			appropriately to prevent the intake of remaining sub therapeutic doses of the drug, which may lead to resistance.
		XI.	Interaction with other medicinal product: When combined with tetracyclines and macrolide antibiotics, enrofloxacin may produce an antagonistic effect. Resorption of enrofloxacin may be reduced, when it is combined with substances containing magnesium or aluminium. The serum level of theophylline may increase when it is combined with enrofloxacin. Do not combine enrofloxacin with non-steroidal antiplogistics. Increased influx of the air (admixing CO ₂ from the air) into medicated drinking water may result in precipitation of enrofloxacin. Precipitation of the salt of enrofloxacin and alkalis may occur at higher concentration of calcium and magnesium in the water system during intermediate dilution in the dosage devices.
		XII.	Pregnancy and Lactation: Laying birds: Do not use in replacement birds within 14 days of coming into lay.
		XIII.	Side Effects: Locomotion disturbances as a result of damage of joint cartilage could not be excluded in case that fluoroquinolones are used during the growing period, particularly at higher temperatures, when consumption of medicated water is drastically increased for longer period.
		XIV.	Symptoms and Treatment of Overdose: Due to the low toxicity of enrofloxacin the danger of overdosing is limited. In cases of significant overdose, transient reduction of mobility and cramps may occur. Symptomatic treatment is recommended in such cases.
		XV.	Withdrawal Period: Chicken: Meat and offal: 7 days Turkey : Meat and offal: 13 days Not authorized for use in lying birds producing eggs for human consumption. Do not administer to layer replacement birds within 14 days of coming into lay.
		XVI.	Storage Condition: Stored at ≤ 30°C / store below 30°C.
		XVII.	Proposed Shelf Life: As packaged for sale: 3 years After first opening of the immediate packaging: 28 days After dilution or reconstitution: 24 hours
		XVIII.	Dosage Forms and packaging available: In liquid form, packed in 1L container.
		XIX.	Name and Address of manufacturer NAM PHARMA SDN. BHD. 5, Lebuhr Perusahaan Klebang 11, Taman Perindustrian Antarabangsa IGB, 31200 Chemor, Perak, Malaysia.
		XX.	Name and Address Product holder KTY CHEMICALS SDN BHD 65, Jalan Cenderawasih 3, Taman Cenderawasih Indah, 14300 Nibong Tebal, SPS, Penang, Malaysia.
		XXI.	Date of Revision of Package Insert: 30/09/2019